

PINNs for COVID-19 in India — Unified Phase-1 Report



Three physics-informed neural network models of COVID-19 spread in India were developed in parallel: the Unified SIRD-PINN, the Domain-Decomposed SEIR-PINN and the RK4-SEIRD model. This report consolidates their approaches, figures and results, and compares them.

1. Problem and shared setup

All three models attack the same inverse problem: recover hidden epidemic rates (transmission β , recovery γ , mortality μ , incubation σ) from noisy Indian COVID-19 data, instead of assuming them constant. A PINN does this by minimising a data-fit loss and the residual of a governing ODE at the same time; because both share the network weights, the rates that come out must jointly explain the data and obey the epidemic model.

Data. Google COVID-19 Open Data joined with Oxford stringency and Google/Apple mobility indices (india_processed_final.csv). It covers roughly 550–597 districts, 991 days, with district centroids (x, y), normalised time t , compartment fractions and covariates (stringency, mobility, population density, vaccination, testing, lagged infections).

Models used. SIRD (Unified SIRD-PINN and the intermediate RK4-SEIRD iterations), SEIR (Domain-Decomposed SEIR-PINN) and SEIRD (final RK4-SEIRD model).

SIRD, with $S + I + R + D = 1$:

$$dS/dt = -\beta SI \cdot dI/dt = \beta SI - \gamma I - \mu I \cdot dR/dt = \gamma I \cdot dD/dt = \mu I$$

SEIR adds an unobserved Exposed compartment with incubation rate σ (deaths folded into R):

$$dS/dt = -\beta SI \cdot dE/dt = \beta SI - \sigma E \cdot dI/dt = \sigma E - \gamma I \cdot dR/dt = \gamma I$$

SEIRD combines both: an Exposed stage plus a separate Deceased compartment driven by μ .

The common acid test. Every model was finally tested by forward-integrating the learned rates with a Runge-Kutta 4 (RK4) solver, independent of the state network. If the rates are physically meaningful, the rollout should reproduce the observed curves. This test is the thread that ties the three models together.

2. Unified SIRD-PINN (district-level)

The Unified SIRD-PINN predicts held-out districts' S, I, R, D with R^2 of 0.97 / 0.79 / 0.96 / 0.98 and exact closure (error 2.4×10^{-7}), but its discovered rates diverge under RK4 rollout. It was trained on 587,663 rows across 593 districts.

2.1 Data pipeline

1. Collapsed 3,964 duplicate (district, t) rows of 591,627 by averaging.
2. Repaired reporting-dropout timesteps (cross-district mean of I below 40% of a 9-step rolling median) by per-district linear interpolation.
3. Regenerated lag features `I_lag_7` and `I_lag_14` after the repair.
4. Ran an automated quality scan. It caught two silent bugs: `stringency_index` is zero in every row (the real signal is `stringency_index_alternate`), and `growth_rate` contains inf values.
5. Split by district, not by row: 474 train / 119 validation districts (469,734 / 117,929 rows). This was the single most important fix.
6. Built a k-nearest-neighbour ($k = 6$) district graph for a mobility-weighted "neighbour infectious pressure" feature.
7. Fit feature scaling on the training split only.

2.2 Earlier two-network prototype and what went wrong

The first design used a TrackerNet, $(x, y, t) \rightarrow (S, I, R, D)$ with independent sigmoids (no sum-to-one guarantee), and an EnvironmentNet producing $(\beta, \gamma, D_{\text{diffusion}}, \mu)$ through an unbounded exponential. They were coupled by a Laplacian diffusion term $D \cdot (I_{xx} + I_{yy})$, which treats infection like heat in a metal plate. That has no epidemiological basis: district centroids are unordered points, and transmission follows mobility between specific neighbours.

- **Data leakage.** Validation used a shuffled row split (50,000 / 15,000). The model trained on day $t = 0.40$ of a district and was tested on $t = 0.41$ of the same district, so every accuracy number was unreliable.
- **Flat-line predictions.** Under an honest district split, the plain sigmoid MLP smoothed away both epidemic waves (spectral bias; Figure 1).
- **Rollout blow-up.** RK4 integration of the rates collapsed I to zero or pushed every compartment to 1.0 within a few steps. It was patched with sub-stepping, hand-picked rate clamps, ground-truth re-injection every 14–15 steps and an " R_0 -boost" multiplier —

evidence the rates were not mechanistically trustworthy.

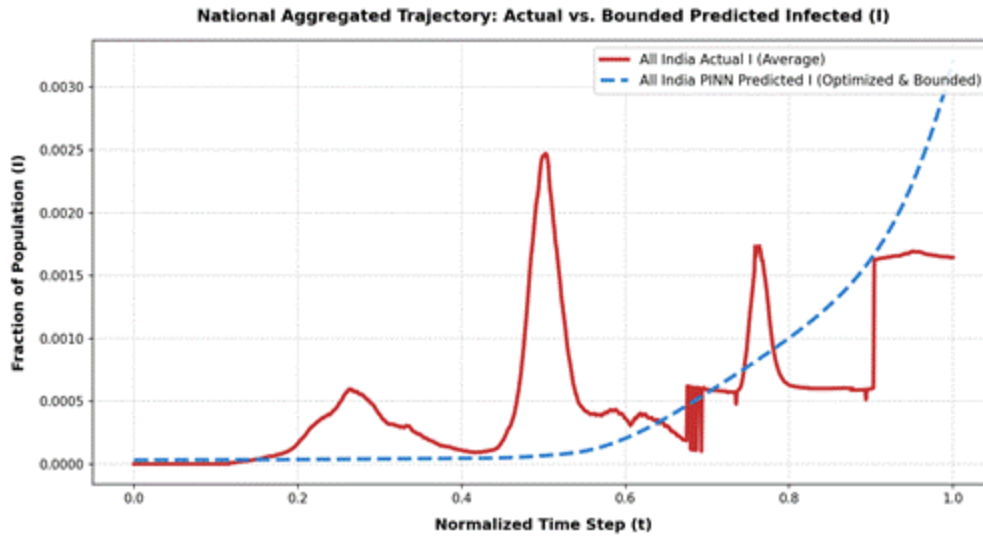


Figure 1. Earlier two-network model: national I , actual vs predicted.

2.3 Final architecture

- **One shared trunk:** 4 fully connected layers $\times$ 128 units, SiLU, dropout 0.1. Input is 9-D: (x, y, t) plus six context features, including neighbour pressure.
- **State head:** Linear $\rightarrow$ SiLU $\rightarrow$ Linear $\rightarrow$ Softmax, so $S + I + R + D = 1$ by construction.
- **Rate head:** Linear $\rightarrow$ Tanh $\rightarrow$ Linear $\rightarrow$ Softplus, giving non-negative (β, γ, μ) without Sigmoid's zero-gradient saturation.
- **Coupling:** the Laplacian is replaced by $\text{diffusion_coupling} \times (\text{neighbour_pressure} - I)$, so pressure flows from sicker neighbours into a district.
- **Loss:** inverse-variance-weighted data loss over all four compartments (so tiny I and D are not drowned by $S \approx 1$), the SIRD residual via automatic differentiation, and a small redundant conservation penalty $|S + I + R + D - 1|$.

Critical fix — time-scaling chain rule. An early run gave γ up to ~ 11 and μ up to ~ 4 per day, implying recovery in hours. Time was normalised to $[0, 1]$ over 991 days, then rescaled to $[-1, 1]$, so autograd returned $dl/d(t_scaled)$ rather than $dl/d(day)$. The network inflated rates by $\sim 990/2 \approx 495\times$ to compensate. Multiplying each time-derivative by $2/990 \approx 0.00202$ restored genuine per-day rates.

Hyperparameter	Value
Optimizer	AdamW (weight decay $1e-5$)

Hyperparameter	Value
Learning rate	0.003, cosine-annealed
Batch size	4,096
Epochs	60 (~48 min, CPU, fixed seed)
Gradient clip	max_norm 1.0
Physics / coupling weight	0.05 / 0.05
k (neighbour graph)	6
Train / val districts	474 / 119

2.4 Results

Total loss fell from ~19,425 at epoch 1 to 0.146 at epoch 60; the physics residual settled near 6.4×10^{-4} within a few epochs (Figure 2). On the 119 unseen districts, direct predictions capture both waves with correct shape and timing, though the second-wave peak is over-predicted (Figure 3).

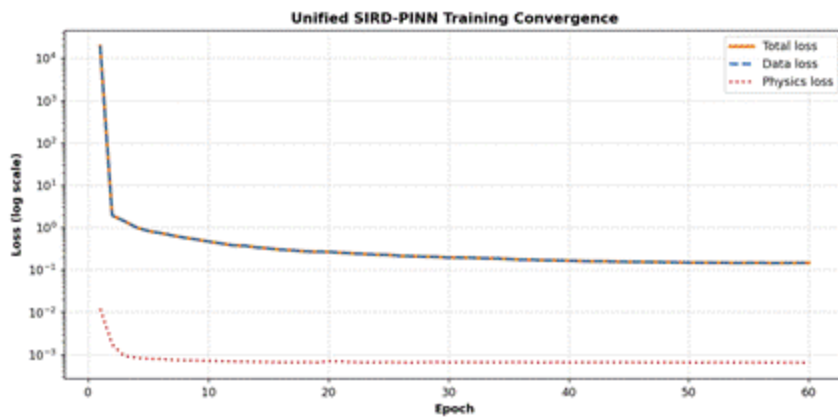


Figure 2. Training convergence (log scale).

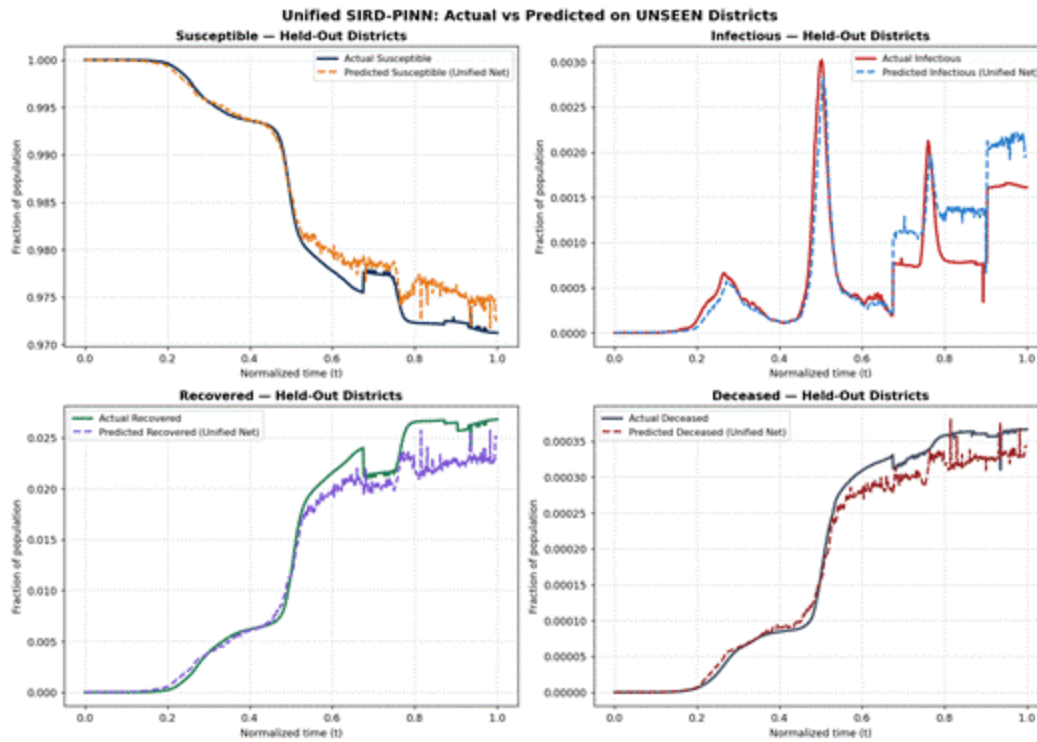


Figure 3. Actual vs predicted S , I , R , D on held-out districts.

Forward RK4 test — the key limitation. Integrating SIRD from the initial condition using only the discovered $\beta(t)$, $\gamma(t)$, $\mu(t)$ diverges almost at once: S collapses to zero and I saturates to 1 within $\sim 15\%$ of the horizon (Figure 4).

Compartment	Direct state-head R^2	RK4-simulated R^2
S	+0.97	-6,396
I	+0.79	-2,118,544
R	+0.96	-341
D	+0.98	-1,424,463

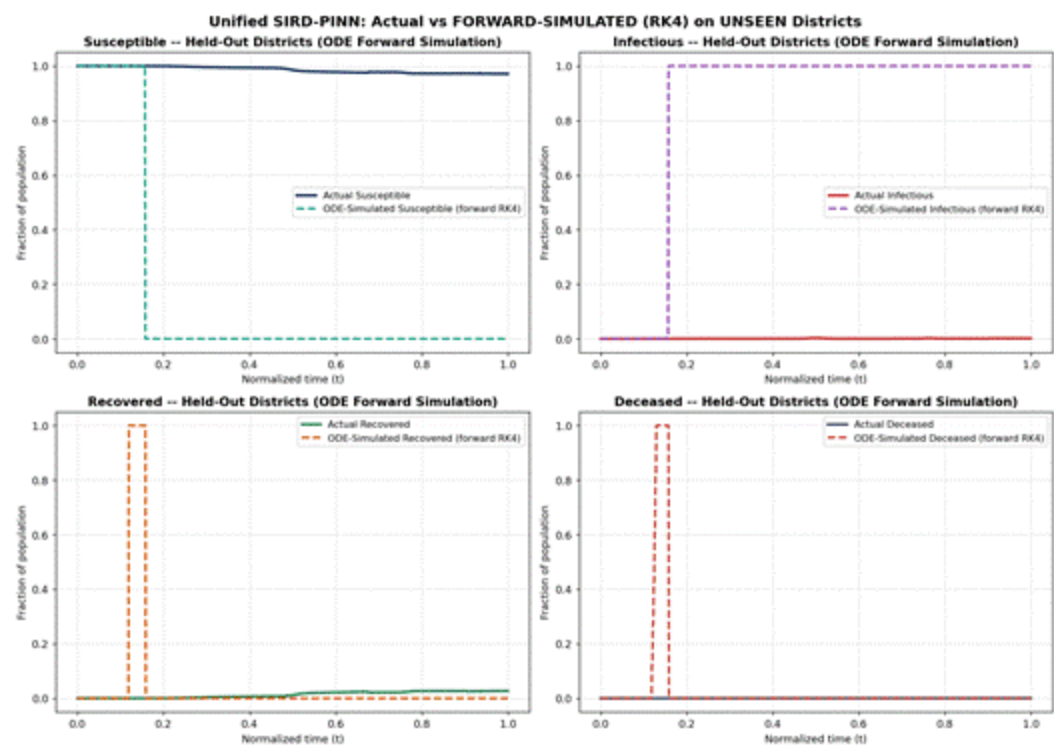


Figure 4. Forward RK4 rollout from the discovered rates alone.

Multi-tier evaluation (held-out districts).

Metric	Earlier model	Unified SIRD-PINN
Validation R ² (I, direct)	not measured leak-free	0.6622
Added Skill vs persistence (5-fold)	not computed	+0.0147 ($\sigma = 0.0285$; 3/5 folds positive)
MSPR, I / D	not computed leak-free	$3.75 \times 10^{-4} / 1.07 \times 10^{-6}$
SIRD closure error	soft penalty only	2.38×10^{-7}

The skill score is only marginally positive and negative in 2 of 5 folds, so the model does not decisively beat naive persistence. The β -vs-stringency correlation is undefined nationally, because averaging collapses stringency's district range (36–74) to near-zero variance.

2.5 Limitations

- Rates reproduce data when read directly but are not yet a standalone mechanistic simulator.
- Marginal forecast skill (+0.015 average).
- β -policy link needs a district- or state-level test.
- No regularisation ties adjacent districts' rates together.
- Single 60-epoch CPU run; no repeated-seed variance or GPU run.

3. Domain-Decomposed SEIR-PINN (per district)

The Domain-Decomposed SEIR-PINN fits the Ernakulam (IN_KL_ERN) district almost perfectly through its state network, and its RK4 rollout follows the timing and shape of the I waves — the closest any residual-based model came to a working mechanistic rollout. The learned rates, however, are often implausibly large and jump between 30-day domains.

3.1 Design

Each district's timeline is split into ~30-day sub-domains ($N_DOMAINS \approx T_DAYS / 30$). Every domain gets two networks:

- **StateNet:** $(t, x, y) \rightarrow (S, E, I, R)$. Linear(3 $\rightarrow$ 64) $\rightarrow$ Tanh $\rightarrow$ 64 $\rightarrow$ 128 $\rightarrow$ Tanh $\rightarrow$ 128 $\rightarrow$ 128 $\rightarrow$ Tanh $\rightarrow$ 128 $\rightarrow$ 64 $\rightarrow$ Tanh $\rightarrow$ 64 $\rightarrow$ 4 $\rightarrow$ Sigmoid. E is never observed; it is inferred purely from the physics residual.
- **ParamNet:** 8 context features $\rightarrow (\beta, \sigma, \gamma)$. Linear(n $\rightarrow$ 64) $\rightarrow$ Tanh $\rightarrow$ Linear(64 $\rightarrow$ 3) $\rightarrow$ Softplus, scaled by $\times 2.0, \times 0.5, \times 0.5$.

Adjacent domains overlap by 10%; a boundary loss makes the piecewise trajectory continuous. Domains are assigned round-robin to available GPUs.

3.2 Data pipeline

Stage	What it does
Cleanup	Drops all-NaN and constant columns
State filter	Keeps states with $\geq 10,000$ rows

Stage	What it does
Encoding	lat/long $\rightarrow$ x, y in $[-1, 1]$; date $\rightarrow$ t in $[0, 1]$; T_DAYS = real span
District slice	aggregation_level = 2, sorted by location and date
Pruning	Drops wikidata_id, new_* daily columns and age columns
Density	population / area, median-imputed
Policy	Five Oxford ordinal closures summed into sum_of_restrictions
Stringency	Missing district values backfilled from state mean, then global mean
Compartments	R = recovered + deceased; I = confirmed - R (≥ 0); S = population - I - R; all divided by population
Forward-fill	Cumulative counts forward-filled per district
Dynamics features	7-day-smoothed velocity and acceleration of I; wavelet (Symlet-5, 3-level) spike detector; tanh regime signal; sigmoid spike signal
Smoothing	7-day centred median on S, I, R
Split	Per-district chronological 70 / 15 / 15
Scaling	MinMaxScaler fit on train only
Collocation weights	Sampling biased toward high curvature of I (70% curvature, 30% uniform, Gaussian jitter)

3.3 Physics residuals and loss

At up to 8,000 collocation points per domain per epoch, autograd gives $\partial S / \partial t$, $\partial E / \partial t$, $\partial I / \partial t$, $\partial R / \partial t$, rescaled by dt_scale for the per-domain $[-1, 1]$ time. Four ODE residuals form loss_ode; three algebraic checks (population conservation, R-consistency, E-consistency) form loss_alg.

$$L = w_{\text{data}} \cdot L_{\text{data}} + w_{\text{ode}} \cdot L_{\text{ode}} + w_{\text{alg}} \cdot L_{\text{alg}} + w_{\text{boundary}} \cdot L_{\text{boundary}}$$

Data and boundary losses use relative squared error, $(\text{pred} - \text{true})/(\text{true} + \epsilon)$, because I can be $\sim 1e-6$ early in an outbreak. $w_{\text{data}} = w_{\text{boundary}} = 1e5$; w_{ode} and w_{alg} start at 0.

3.4 Training schedule

- Adam with two parameter groups: StateNet LR $1e-3$, ParamNet LR $1e-2$, each clipped separately at $\text{max_norm } 1.0$.
- Warm-up for 1,000 epochs: physics weights 0 and ParamNet frozen, so StateNet first fits the data.
- At epoch 1,000, physics weights jump to 1.0, then ramp linearly to $1e5$ over 1,000 epochs.
- Checkpoint and log every 500 epochs, to avoid GPU→CPU sync and disk bottlenecks.
- Post-training diagnostics: per-domain rate statistics, StateNet-only fit error and stitched fit plots, then an independent RK4 rollout.

3.5 Debugging history — the 13 modifications

Mod	Problem	Fix
1-4	Colab → Drive → Kaggle migration	Re-routed data and checkpoints
5	fp16 AMP overflowed (targets $\sim 1e-6$, weights $\sim 1e5$); GradScaler silently skipped every optimizer step	Train fully in fp32
6	Shared LR/clip starved ParamNet	Two optimizer groups
7	One shared ParamNet across domains	One ParamNet per domain (more expressive, loses cross-domain regularisation)
8	RK4 loss weight jumped $0 \rightarrow 1e5$ as ParamNet unfroze; training stalled at epoch 1,000	Linear ramp of RK4 weight
9	RK4 loss term in training	Removed entirely; RK4 kept only as post-hoc evaluation
10	Multi-GPU device mismatch in RK4 rollout when crossing domains	Build input tensor on the selected ParamNet's own device

Mod	Problem	Fix
11	ParamNet gradients stayed below clip threshold; it barely trained	Ramp ode/alg weight from 1.0 to 1e5
12	No way to localise faults	Post-training diagnostics (fit vs rollout)
13	Sigmoid saturation pinned β , γ at 0 or range max	Sigmoid $\rightarrow$ Softplus

Hyperparameter	Value
Sub-domain length	30 days
Epochs	7,000 (all domains jointly)
Warm-up	1,000 epochs
ode/alg ramp	1,000 epochs to 1e5
Overlap	10%
Collocation max	8,000 per domain per epoch (70% curvature-biased)
LR StateNet / ParamNet	1e-3 / 1e-2
Initial weights	data = boundary = 1e5; ode = alg = 0
Gradient clip	1.0 per group

3.6 Results — Ernakulam (IN_KL_ERN)

Top rows of each figure show the StateNet direct fit; bottom rows show the RK4 rollout driven only by ParamNet on the training window ($t \leq 0.7$). StateNet tracks S, I, R closely. The rollout reproduces the I peaks' timing but under-predicts depletion: S ends near 0.925 against a true ~ 0.81 , and R near 0.09 against ~ 0.185 . Results at 5,000 and 7,000 epochs are almost identical, so training had converged.

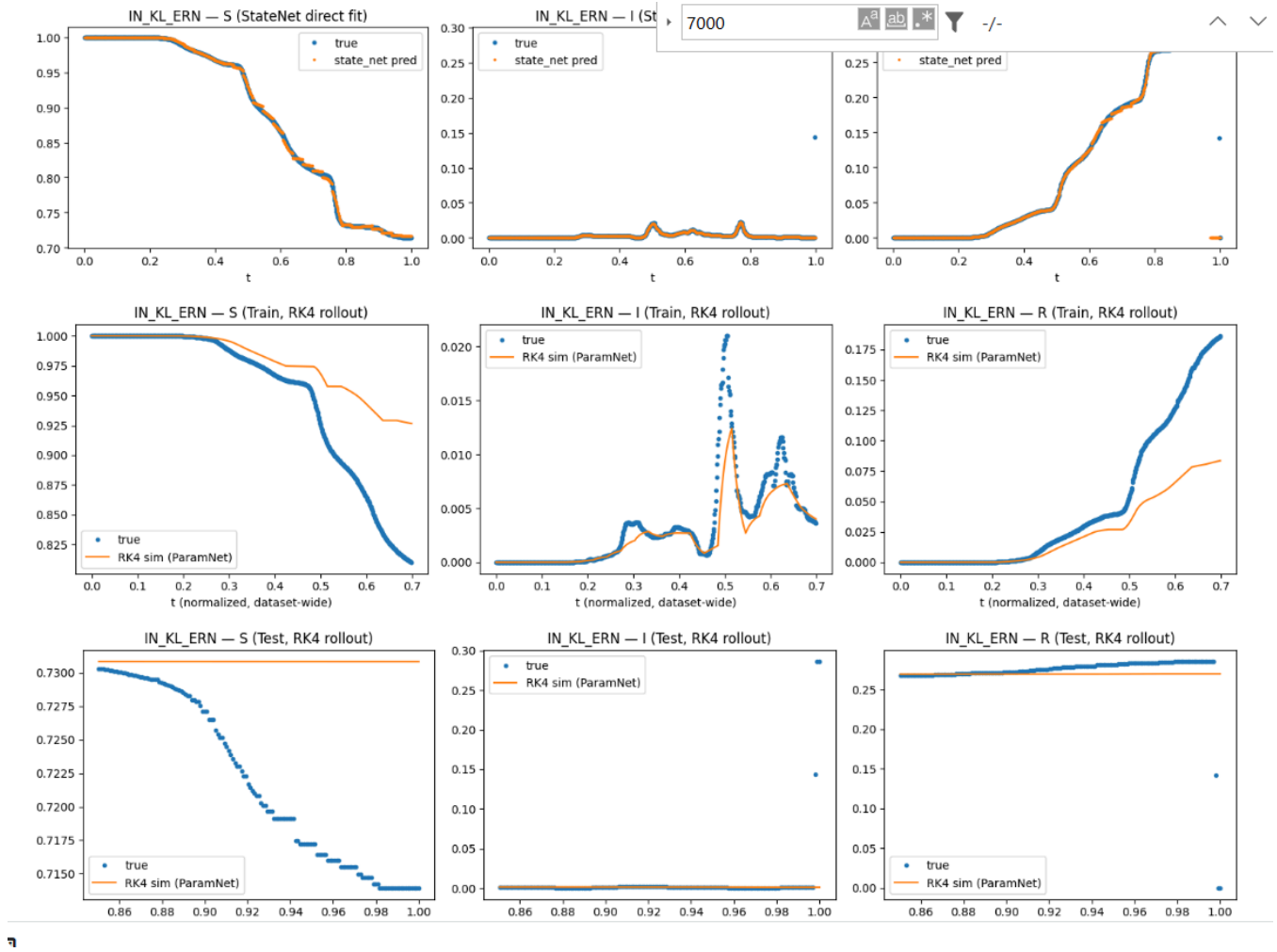


Figure 5. IN_KL_ERN after 5,000 epochs: StateNet fit (top) and RK4 rollout (bottom).

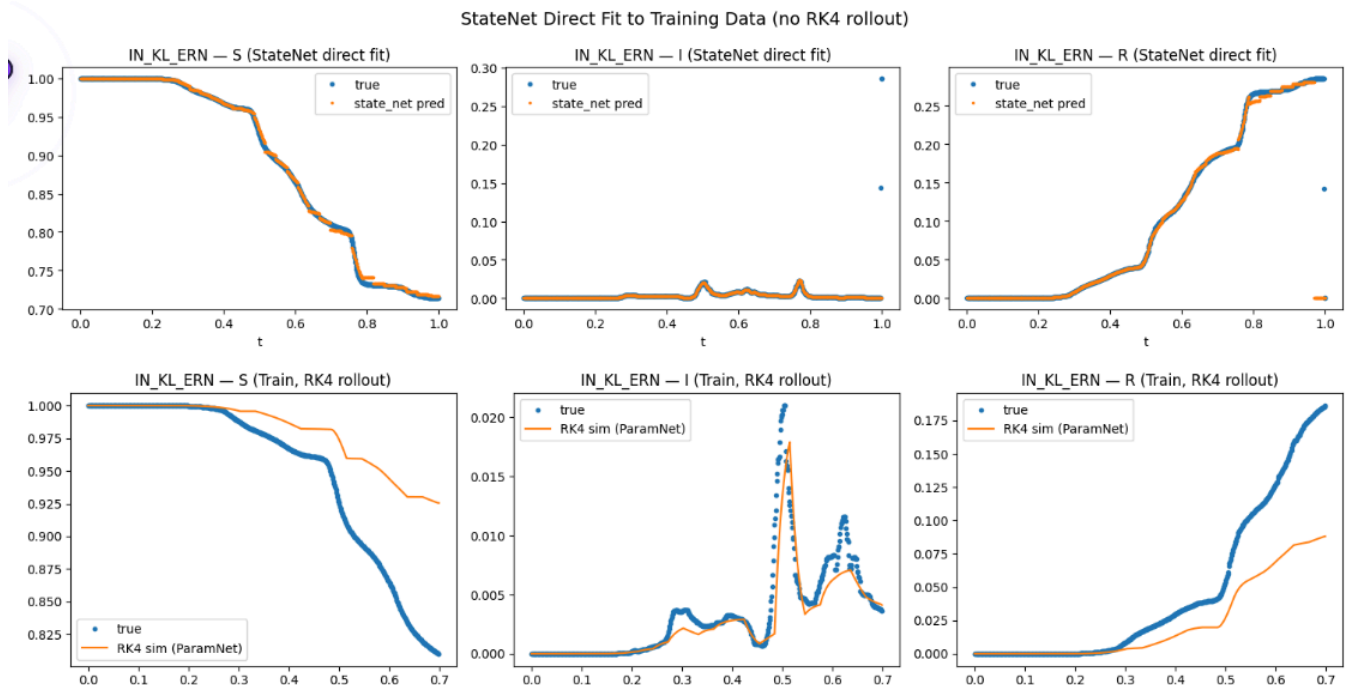


Figure 6. IN_KL_ERN after 7,000 epochs.

Learned parameters per domain (IN_KL_ERN, 31 rows each). Within-domain standard deviation is ~ 0 almost everywhere, meaning ParamNet outputs a near-constant per domain despite varying inputs. β reaches 68 per day and γ 62 per day, far outside plausible epidemiology ($\beta \sim 0.1$ – 1 , $\gamma \sim 0.1$), and values jump between neighbouring domains — a consequence of missing cross-domain regularisation.

Domain	β	σ	γ
0	0.0070	0.0002	0.0007
1	0.0096	0.0000	0.0059
2	2.2157	0.0000	0.2214
3	15.0413	0.0000	0.0013
4	14.4980	0.0002	2.9641
5	31.8896	0.0013	0.0361
6	44.2666	0.0064	13.7004
7	55.4512	0.0211	40.9176
8	49.0512	0.0536	44.0214
9	46.5122	0.1137	43.9232
10	0.0001	0.0003	9.4542
11	61.1017	0.1338	61.0022
12	59.0463	0.1677	57.2690
13	63.0585	0.1830	62.0935
14	2.7542	0.0034	42.6394
15	9.3669	0.0274	0.0002
16	62.2587	1.2327	56.2942
17	1.3769 ± 0.0188	0.0039	55.9133 ± 0.0728
18	60.6024	0.2368	53.4497
19	67.8063	0.4107	58.2683

Domain	β	σ	γ
20	59.4927 ± 0.0212	0.3714	49.9644 ± 0.0220
21	0.0743 ± 0.0114	0.0002	12.6925 ± 0.0177
22	36.4473	0.1210	31.3726
23	9.4683	0.0289	14.7847
24	4.8045	0.0243	0.0005
25	34.7038 ± 0.2761	0.1029 ± 0.0018	27.8689 ± 0.2942
26	0.0022	0.0001	19.4121
27	2.3981	0.0022	4.4776
28	0.0237	0.0000	4.6063
29	0.0001	0.0014	0.0059
30	12.7993	0.0189	9.8664
31	0.0352	0.0000	1.7257
32	0.0080	0.0005	0.0047

Standard deviations are 0.0000 unless shown. In many busy domains $\beta \approx \gamma$ with both large, so their ratio (the effective reproduction number) stays near 1 — the fit balances two big rates rather than finding small realistic ones.

3.7 Open issue

Per-domain ParamNets have no soft-tying loss at domain edges, so β , σ , γ can jump discontinuously even though the state trajectory is constrained to be continuous.

4. RK4-SEIRD — from plain SIR to a differentiable-RK4 SEIRD (national)

The final RK4-SEIRD model abandons the residual-PINN formulation: a network predicts time-varying β , σ , γ , μ , and a differentiable RK4 solver produces the trajectory, so every predicted curve satisfies SEIRD exactly. It cut loss by ~95.5% ($6.3772 \rightarrow 0.28945$) and tracks the national waves, at the cost of district resolution. It got there through four iterations, each answering the previous one's failure.

4.1 Iteration 1 — Plain SIR inverse PINN

β and γ are two global trainable scalars; the network maps only $(x, y, t) \rightarrow (S, I, R)$, with no covariates. Physics weight was raised from 1% to 20% of the data loss, and 10,000 full-batch epochs worked far better than 2,000.

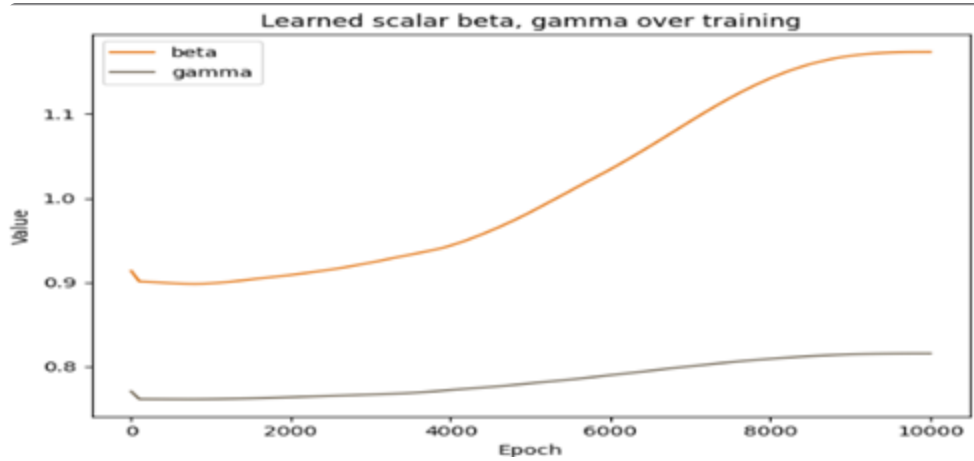


Figure 7. Iteration 1: learned β rises from ~ 0.90 to ~ 1.17 ; γ stays near 0.76 – 0.82 .

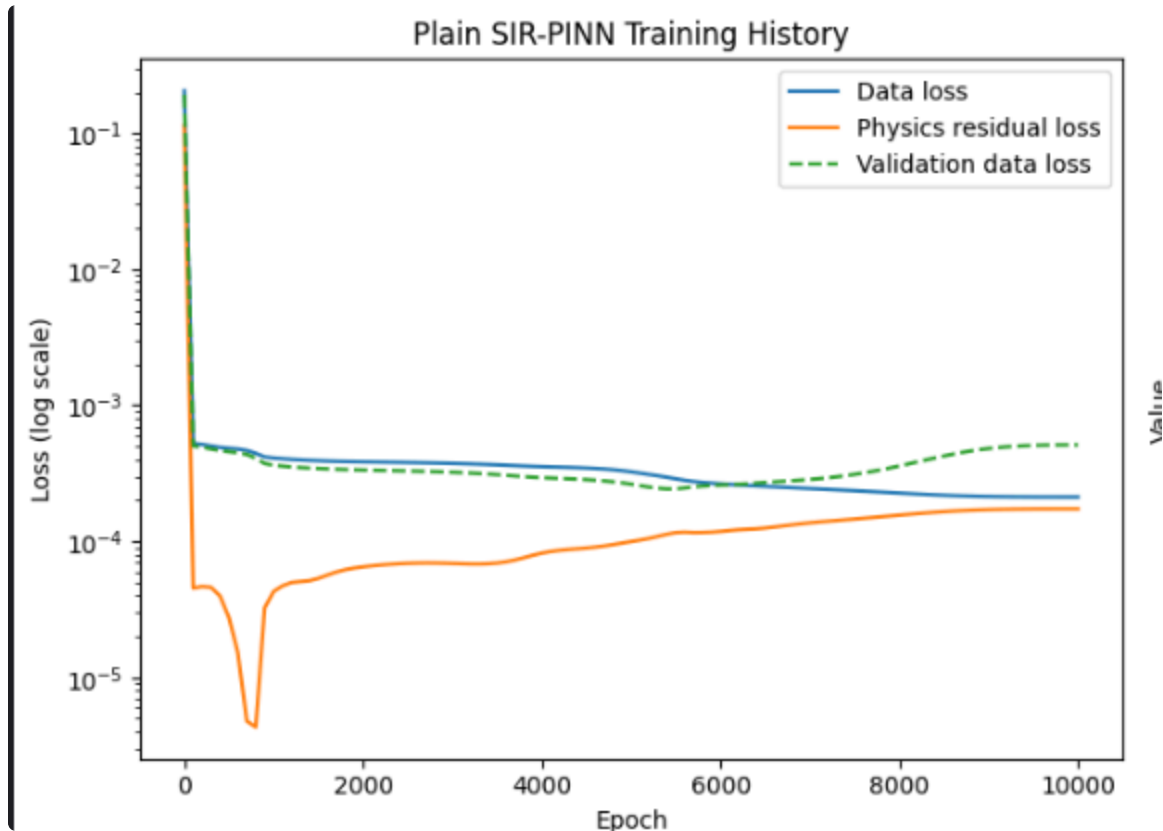


Figure 8. Iteration 1 training history: validation loss starts rising after $\sim 5,500$ epochs while training loss keeps falling (overfitting).

Problems: ~ 550 districts $\times$ 991 days squeezed into one β and one γ gives a national-average compromise; the physics loss was small next to the data loss, so the model fit data greedily; γ barely moved; no peaks were captured.

4.2 Iteration 2 — SIR with Fourier features and peak weighting

- **Spectral bias:** one district fits to MSE $\sim 6e-6$, but many districts through one (x, y, t) function stall at a flat average. Fix: Fourier features on time.
- **Loss imbalance:** $\sim 69\%$ of (district, t) rows have $I < 0.0001$ and only $\sim 9\%$ have $I > 0.001$. Fix: data loss weighted by I relative to each district's own peak, so every district's peak counts equally.
- **Physics as regulariser:** with physics off, the network fit the first peak but oscillated up to $4\times$ the true peak elsewhere. Physics stays on at $\lambda = 1.0$ (an earlier 0.001 day-scaling compensation was wrong and removed).
- **New evaluation:** solve the plain SIR ODE with RK4 using the learned β, γ , to check they behave as real SIR parameters.

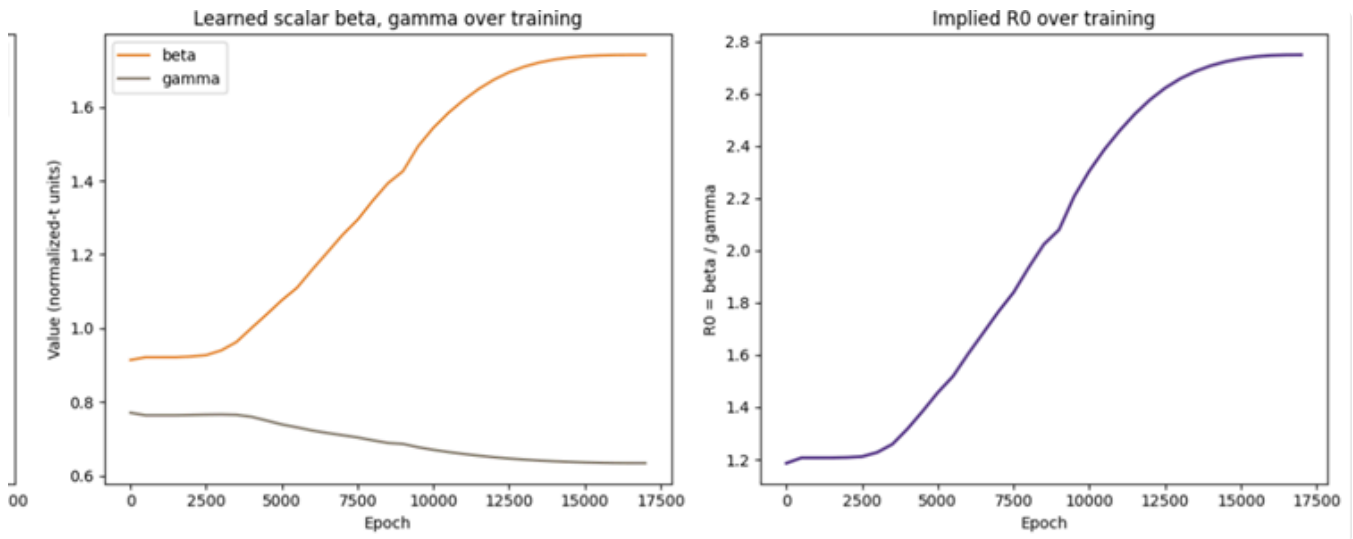


Figure 9. Iteration 2: β rises to ~ 1.75 , γ falls to ~ 0.63 , implied $R_0 = \beta/\gamma$ rises from ~ 1.2 to ~ 2.75 (normalised-time units).

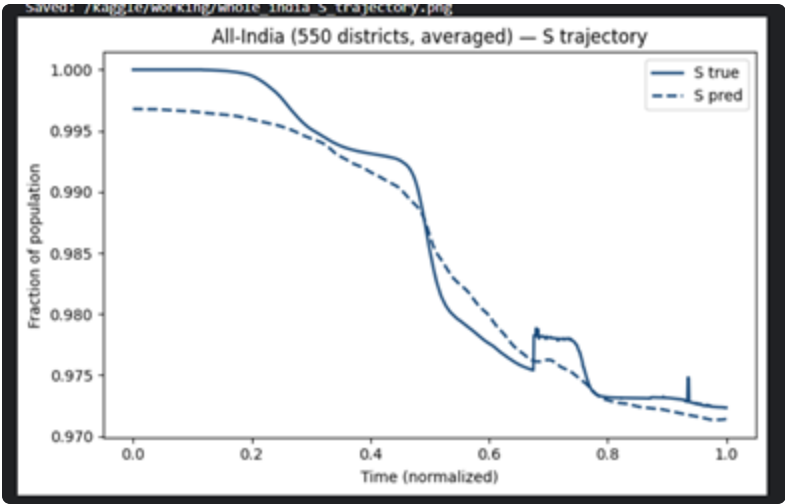


Figure 10. All-India (550 districts averaged) S: the trend is followed but steps are smoothed.

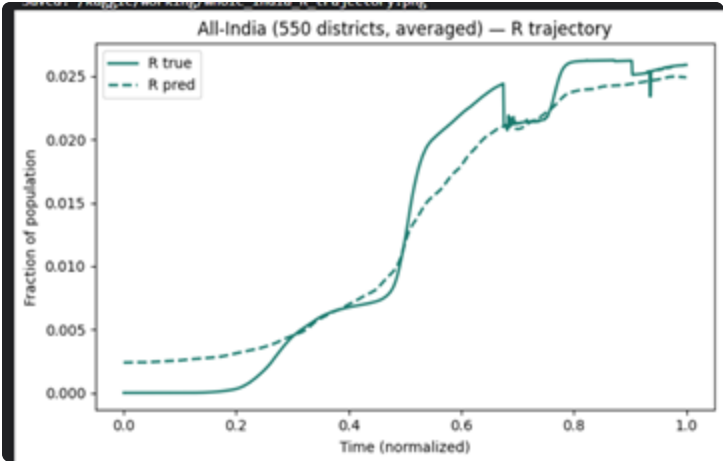


Figure 11. All-India R: close overall, with an offset at $t = 0$.

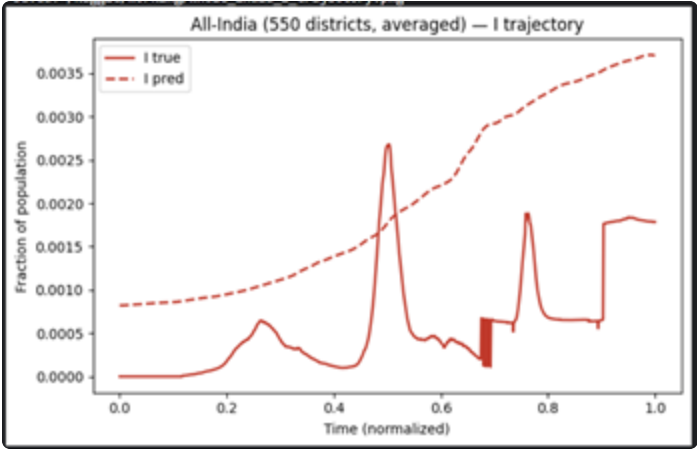


Figure 12. All-India I: prediction rises monotonically and misses every wave.

Hyperparameter	Value
Hidden layers $\times$ units	7×128 , Tanh
Time Fourier frequencies / scale	24 / 6.0
Input $\rightarrow$ output	$(x, y, t) \rightarrow (S, I, R)$
β, γ	2 global trainable scalars
Collocation points	20,000
Train / val split	85% / 15% of districts
Data / physics / I-residual weight	10 / 1 / 10
Epochs	17,000
Optimizer	Adam, LR $1e-3$, cosine annealing, seed 42

Problems: γ still nearly flat, and the RK4 solution from the learned β, γ did not reproduce the data.

4.3 Iteration 3 — Spatiotemporal SIRD with covariate-driven rates

SIR becomes SIRD, and β, γ, μ become functions of (x, y, t) and all covariates. A key refinement: Fourier-encoding (x, y) as well as t made things worse at 80 districts, causing ringing between districts, because (x, y) act as fixed per-district IDs rather than a continuous field. The fix was to Fourier-encode t only and pass (x, y) raw.

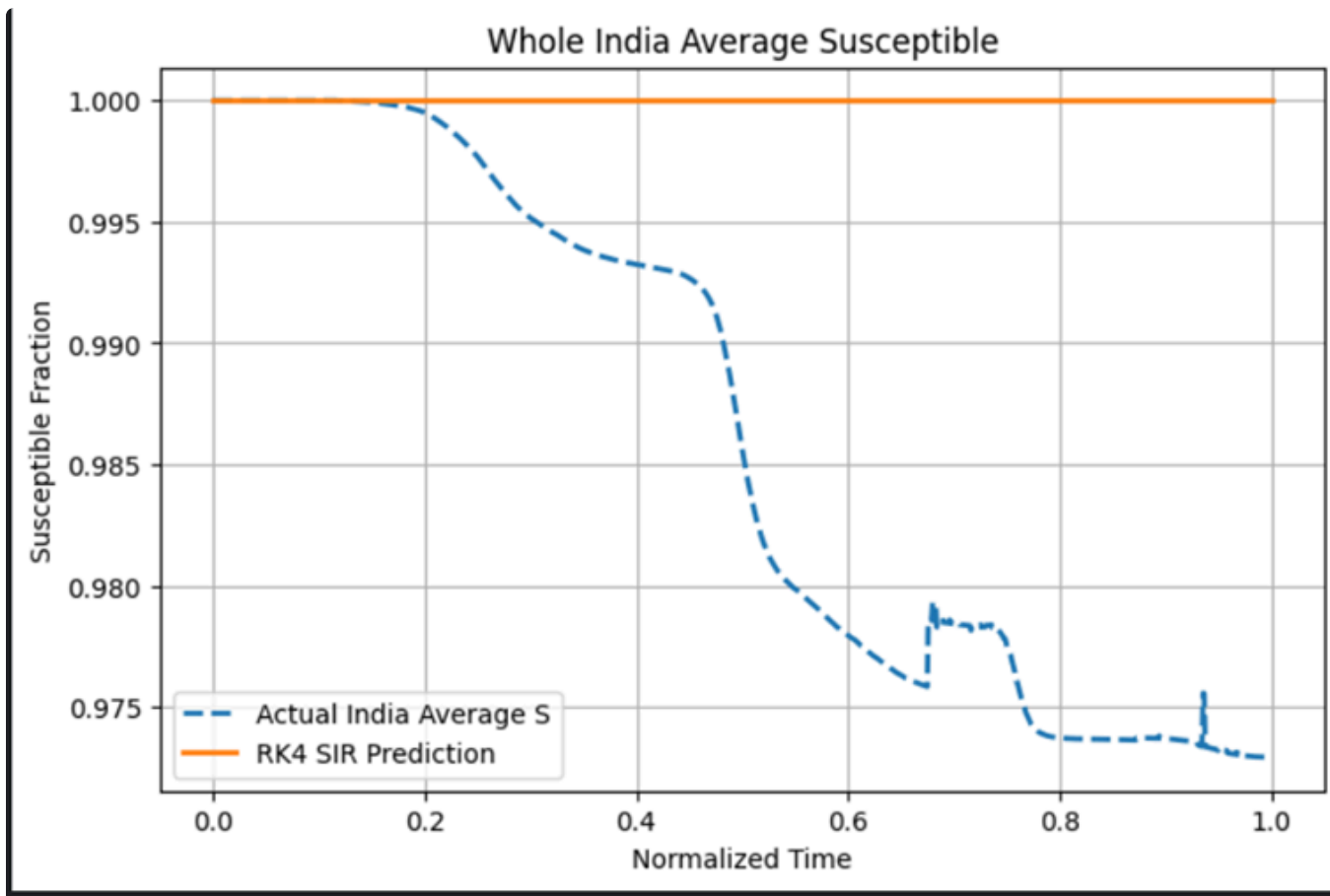


Figure 13. Iteration 3: RK4 rollout keeps S flat at 1.0 while actual S falls to ~ 0.973 — no epidemic is simulated.

4.4 Iteration 4 (final) — SEIRD with differentiable RK4 as the training loss

A network takes time, Fourier features and covariates, and outputs four positive (Softplus) time-varying rates β , σ , γ , μ . These drive SEIRD inside a differentiable PyTorch RK4 integrator; gradients flow back through the solver. The simulated trajectory is compared with data using variance-weighted MSE, plus a smoothness penalty on the rates.

The difference from a standard PINN: there is no state network and no residual loss. The solver enforces the physics exactly, so the model cannot invent a curve that violates SEIRD.

Inputs (26): time t ; 16 Fourier features $\sin(2\pi kt)$, $\cos(2\pi kt)$ for $k = 1 \dots 8$; 9 covariates — stringency, vaccination rate, full vaccination rate, testing rate, tests per case, mobility score, growth rate, $I(t-7)$, $I(t-14)$.

$$L = L_{\text{data}} + 5 \cdot L_{\text{smooth}}$$

$$L_{\text{data}} = \text{MSE}(I)/\text{Var}(I) + \text{MSE}(R)/\text{Var}(R) + \text{MSE}(D)/\text{Var}(D)$$

$$L_{\text{smooth}} = \text{MSE}(\Delta\beta) + \text{MSE}(\Delta\gamma) + \text{MSE}(\Delta\mu)$$

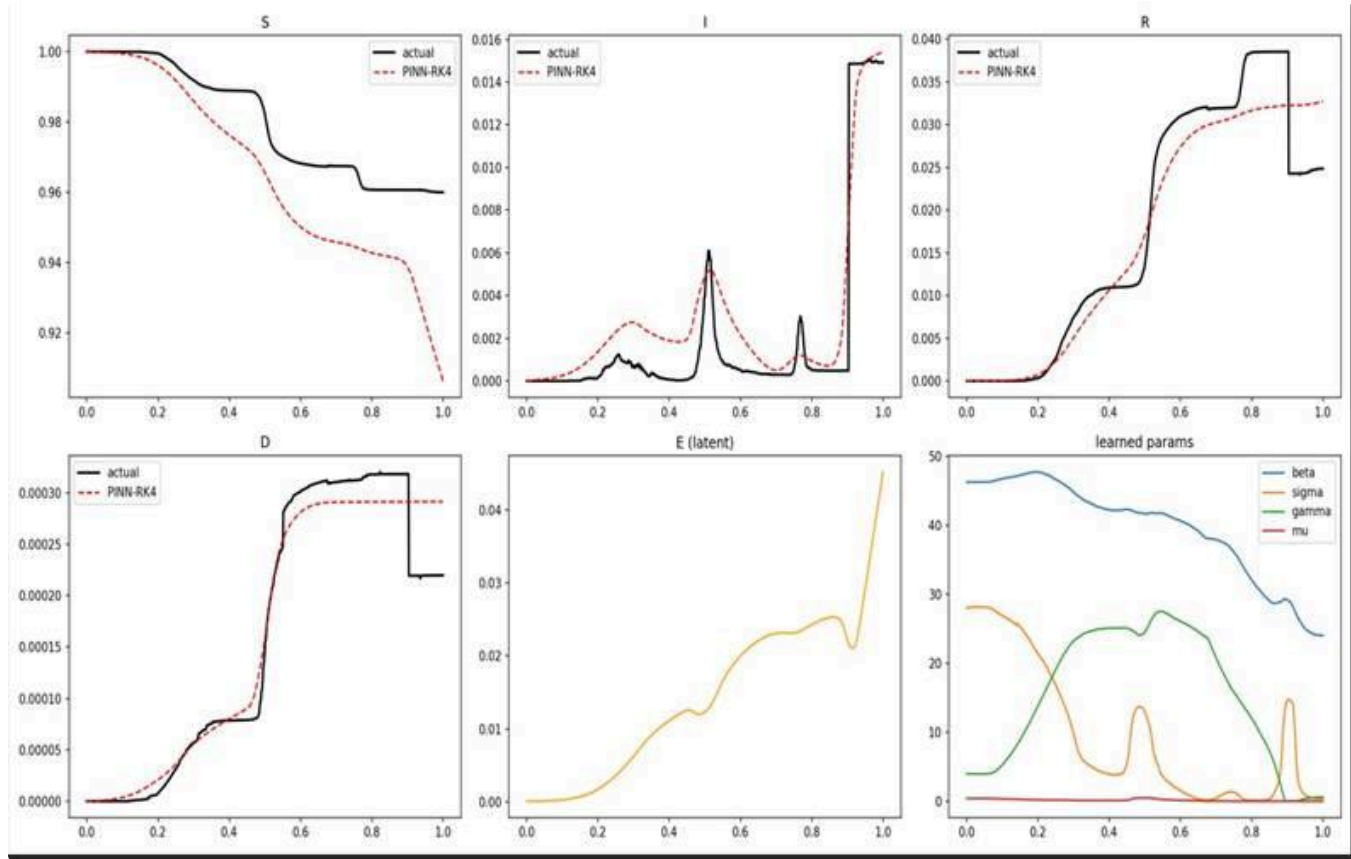


Figure 14. Final model, national aggregate: actual vs PINN-RK4 for S , I , R , D , the latent E , and learned β , σ , γ , μ over time.

Reading Figure 14: I follows both waves and the late surge, with the first wave over-predicted and peaks smoothed. D matches closely until the final data drop. R tracks the rise but not the late jumps. S over-depletes toward the end (≈ 0.905 vs ≈ 0.96). Learned β declines from ~ 46 to ~ 24 , γ rises to ~ 25 mid-period then falls, σ drops early with two spikes, and μ stays near zero. These rates appear to be per unit of normalised time ($dt = t[1] - t[0]$), so they are not directly per-day values.

Category	Setting	Value
Model	MLP	$26 \rightarrow 2 \times 64 (\text{Tanh}) \rightarrow 4$
Rates	β	$\text{softplus}(o_0) \times 1.2$
Rates	σ	$\text{softplus}(o_1) \times 0.5 + 0.05$
Rates	γ	$\text{softplus}(o_2) \times 0.3 + 0.02$

Category	Setting	Value
Rates	μ	$\text{softplus}(o_3) \times 0.02$
Initial condition	E_0	$\text{softplus}(-6.0) \times 0.01$
Solver	RK4	$dt = t[1] - t[0]$, states clipped to $[0, 1.2]$
Optimizer	Adam	LR $2e-3$, StepLR (step 800, γ 0.5), clip 5.0
Training	Epochs	3,000
Loss	Data	1/Var-weighted MSE on I, R, D
Loss	Smoothness	coefficient 5.0 on β , γ , μ
Data	Aggregation	population-density-weighted national average
Data	Covariates	standardised; missing $\rightarrow 0$
Evaluation	Metric	RMSE for S, I, R, D

Training summary	Value
Initial / final total loss	6.3772 / 0.28945
Initial / final data loss	6.3768 / 0.25637
Final smoothness loss	0.006617
Loss reduction	$\sim 95.46\%$
Training time	9,968.2 s (~ 2.77 h)

Note: the Softplus scaling in the table ($\beta \leq \sim 1.2 \times \text{softplus}$) sits oddly with $\beta \approx 46$ in Figure 14; the plotted values may be rescaled. Worth confirming in the notebook.

5. Side-by-side comparison

The three models trade spatial reach against mechanistic faithfulness: the Unified SIRD-PINN covers all districts but its rates fail RK4; the RK4-SEIRD model's rates pass RK4 by construction but only nationally; the Domain-Decomposed SEIR-PINN sits between, per district with a partially working rollout.

Aspect	Unified SIRD-PINN	Domain- Decomposed SEIR- PINN	RK4-SEIRD
Compartments	S, I, R, D	S, E, I, R (deaths in R)	S, E, I, R, D
Spatial scope	593 districts, one model	One district per run (IN_KL_ERN shown)	National population-weighted average
Time handling	Single network over 991 days	~30-day sub-domains, 10% overlap	Single run, Fourier features $k = 1 \dots 8$
Physics enforcement	Residual loss via autograd (soft)	Residual + algebraic + boundary losses (soft)	Differentiable RK4 solver (exact)
State output	Softmax head, sum = 1 exactly	Sigmoid StateNet, no closure guarantee	Produced by the solver
Rate model	Shared trunk, Softplus head: β, γ, μ	Per-domain ParamNet, Softplus: β, σ, γ	MLP, Softplus: $\beta, \sigma, \gamma, \mu$
Spatial coupling	k-NN ($k = 6$) mobility-weighted pressure	None	None
Validation split	474 / 119 districts (leak-free)	Chronological 70/15/15 per district	Not stated for final model
Training	60 epochs, ~48 min CPU	7,000 epochs, multi-GPU	3,000 epochs, ~2.77 h
Headline result	Direct R^2 S/I/R/D 0.97/0.79/0.96/0.98; I R^2 0.66 on held-out	Near-perfect StateNet fit; rollout follows I waves	Loss -95.5%; RK4 trajectory tracks national waves
RK4 rollout	Diverges within ~15% of horizon	Tracks I timing; under-depletes S, R	Is the model — passes by construction

Aspect	Unified SIRD-PINN	Domain- Decomposed SEIR- PINN	RK4-SEIRD
Rate plausibility	Per-day scale after chain-rule fix	β, γ up to $\sim 60/\text{day}$; jumps between domains	Smooth; units need confirming
Main weakness	Rates not mechanistically usable	Implausible, discontinuous rates	No district heterogeneity

6. Shared lessons

The three models hit the same pitfalls independently, which marks them as generic PINN problems rather than implementation bugs.

1. **Direct fit is not physics.** Every state-network model fit the data well while its rates failed forward integration. The Unified SIRD-PINN's R^2 fell from $+0.79$ to -2.1 million for I ; RK4-SEIRD iterations 1–3 all failed the RK4 check. A low residual loss alone does not validate the rates.
2. **Time normalisation breaks the chain rule.** The Unified SIRD-PINN ($\sim 495\times$ inflation) and the SEIR-PINN (dt_scale) both had to rescale autograd derivatives from normalised to real time; the RK4-SEIRD line removed a wrong 0.001 compensation. Unscaled rates look plausible in shape but are orders of magnitude off.
3. **Sigmoid saturates rate heads.** The SEIR-PINN's β, γ pinned at range limits; both it and the Unified SIRD-PINN moved to Softplus.
4. **Spectral bias flattens waves.** The two-network prototype and RK4-SEIRD iteration 1 smoothed away peaks. Fourier features on time helped; Fourier features on (x, y) hurt, because district coordinates are IDs, not a continuous field.
5. **Loss imbalance.** Tiny I and D get drowned by $S \approx 1$. Fixes: inverse-variance weights (Unified SIRD-PINN, RK4-SEIRD), relative error (SEIR-PINN), per-district peak weighting (RK4-SEIRD iteration 2).
6. **Leakage and honest splits.** Row-shuffled splits inflated the prototype's early numbers; district-level or chronological splits are required.
7. **Rates need regularisation across space and time.** The Unified SIRD-PINN has none across districts; the SEIR-PINN has none across domains. RK4-SEIRD's smoothness penalty is the only explicit example, and its rates are the smoothest.

8. **Loss-weight schedules matter.** The SEIR-PINN's stalls and starved ParamNet were fixed by ramps rather than jumps; fp16 silently skipped all updates under $1e5$ -scale weights.

7. Conclusion

Phase 1 shows that PINNs can fit Indian COVID-19 curves well, but only an architecture that enforces the ODE exactly yields rates that reproduce the epidemic when integrated. Soft residual losses let the state network absorb the data while the rates drift.

Each approach contributes a piece the others lack:

- **Unified SIRD-PINN:** the strongest data engineering and evaluation — leak-free district split, quality scan, closure guarantee, mobility-based coupling, persistence baseline and an honest RK4 test.
- **Domain-Decomposed SEIR-PINN:** domain decomposition and curvature-biased collocation that capture sharp regime changes, plus the most thorough optimisation diagnostics.
- **RK4-SEIRD:** the differentiable-RK4 formulation, the only one whose trajectories obey the physics by construction, with smooth rate curves.

Recommended next phase. Combine them: the RK4-SEIRD differentiable-solver core, run per district on the Unified SIRD-PINN's leak-free pipeline with its k-NN mobility coupling inside the ODE, and the SEIR-PINN's decomposition as time windows linked by continuity losses on both states and rates.

1. Add a rollout-consistency loss (or make RK4 the model) in the district-level setup.
2. Regularise rates across neighbouring districts and adjacent time windows, plus a smoothness penalty.
3. Report rates in per-day units with the chain-rule scaling checked, and sanity-check against known ranges (e.g. recovery 7–14 days).
4. Evaluate every model on the same held-out districts with the same metrics: RK4 R^2 , Added Skill vs persistence, and closure error.
5. Recompute β -vs-stringency correlation at district or state level to test the policy-sensitivity claim.
6. Replace the k-NN graph with real mobility-flow data, and run repeated seeds on GPU for variance estimates.